

STAT 240

Week 12: HTML

Owen G. Ward

Today

- Quick Review of Last Week
 - Only the EM stuff, the example not important
- Extracting Data from HTML
- Review for Coding Exam

Logistics

- Coding Exam next Friday
- Written Exam April 14th
 - Practice problems for this already on canvas
 - All of class next Wednesday will be review for this

Review: Likelihood and Mixture Models

Likelihood

- The likelihood is the “probability” of the data given the parameter
- It is a function of the parameter, $\mathcal{L}(\theta|x)$
- Maximum Likelihood Estimation finds the value of θ which maximises this function $\mathcal{L}(\theta|x)$
 - This will be the same value of θ which maximises $\log \mathcal{L}(\theta|x)$

Likelihood functions

- Suppose we flip a coin several times, $P(Heads) = \theta$
- Get n_H heads and n_T tails, likelihood given by

$$\mathcal{L}(\theta|x) = \theta^{n_H}(1 - \theta)^{n_T}$$

- This is the product of likelihood for each individual coin flip, because independent
- The log-likelihood is

$$\log \mathcal{L}(\theta|x) = n_H \log \theta + n_T \log(1 - \theta)$$

Maximum Likelihood

- Find the value of θ which maximises either of these functions
- Can do this using calculus for this problem, others harder
 - No unique solution in many problems
- Can also use `optim` for problems like this, may not work in harder problems

Mixture Models

- Let's see even simpler example than last week
- Let's suppose we have two normal distributions, $\mathcal{N}(0, 1)$ and $\mathcal{N}(-5, 2)$
- For each data point X_i we flip a coin with $P(Heads) = \theta$
 - If the coin is heads then $X_i \sim \mathcal{N}(0, \sigma = 1)$
 - If the coin is tails then $X_i \sim \mathcal{N}(-5, \sigma = 2)$
- We only observe X_i , not each coin flip

Simulating Data for this Model

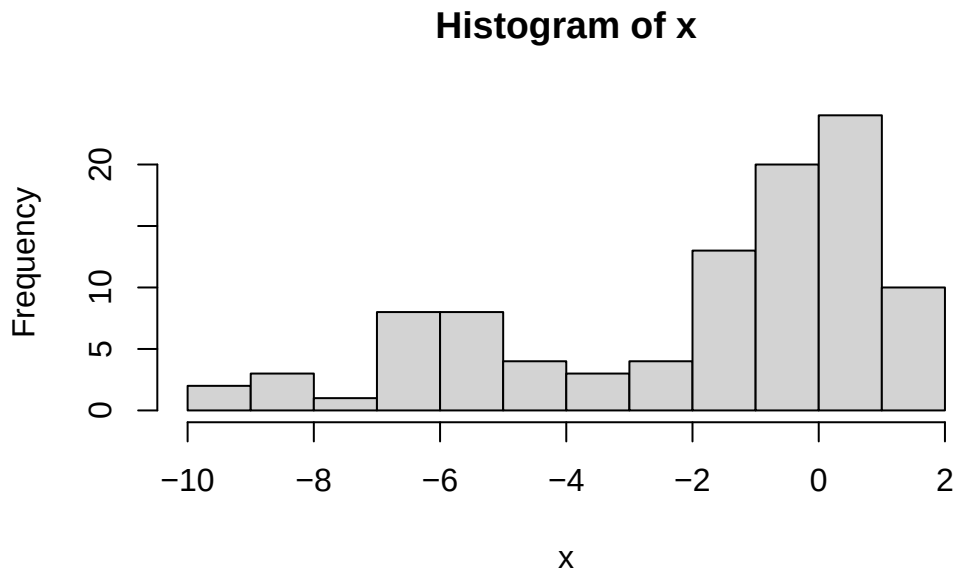
```
set.seed(240)
theta <- 0.7
x <- rep(NA, 100)
coins <- sample(1:2, size = 100, replace = TRUE, prob = c(theta, 1 - theta))
coins[1:10]
```

```
[1] 1 2 1 1 1 1 1 1 1 1
```

```
for(i in 1:length(coins)) {
  if(coins[i] == 1) {
    x[i] <- rnorm(1, mean = 0, sd = 1)
  }
  if(coins[i] == 2) {
    x[i] <- rnorm(1, mean = -5, sd = 2)
  }
}
```

The Data

```
hist(x)
```



Exercise

- Simulate some new data where we make the σ for both normal distributions smaller. What happens to the histogram?

Solution

Fitting a Mixture Model

- The likelihood for this model is (with simplifications)

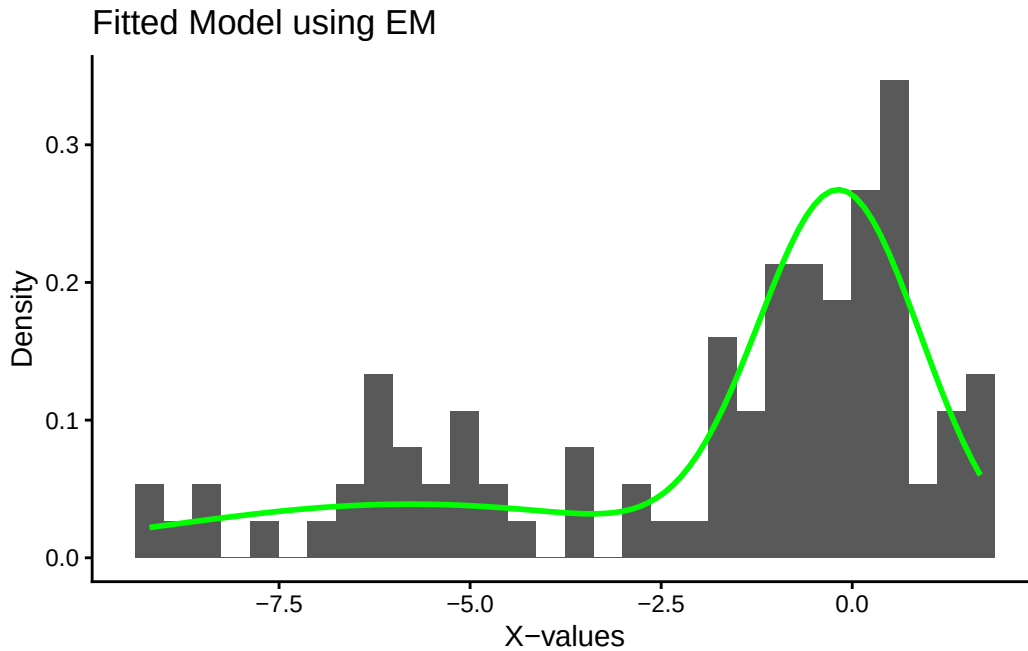
$$\mathcal{L}(\theta|x) = \theta\mathcal{N}(0, 1) + (1 - \theta)\mathcal{N}(-5, 2)$$

- We cannot find the Maximum Likelihood for this model if we didn't know the true Normal distributions

EM Algorithm

```
library(EMcluster)
restarts <- 100
models <- vector("list", restarts)
lls <- rep(NA, restarts)
for (i in 1:restarts) {
  models[[i]] <- emcluster(as.matrix(x),
                           simple.init(as.matrix(x), nclass = 2))
  lls[i] <- models[[i]]$llhdval
}
model <- models[[which.max(lls)]]
```

The Result



More Details on Mixture Models

- This mixture model doesn't classify each point into a cluster, it gives us the probability each data point is in each of the components

```
model$pi
```

```
[1] 0.3119936 0.6880064
```

- So somewhat different to k-means, but useful
- Common in other more advanced Machine Learning methods

HTML

HTML Basics

- HTML is Hypertext markup language
- Best online reference: <https://www.w3schools.com>

HTML: Links

```
<a href="http://www.example.com">A link</a>
```

A link

HTML: Full page

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>This is a Heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

This is a Heading

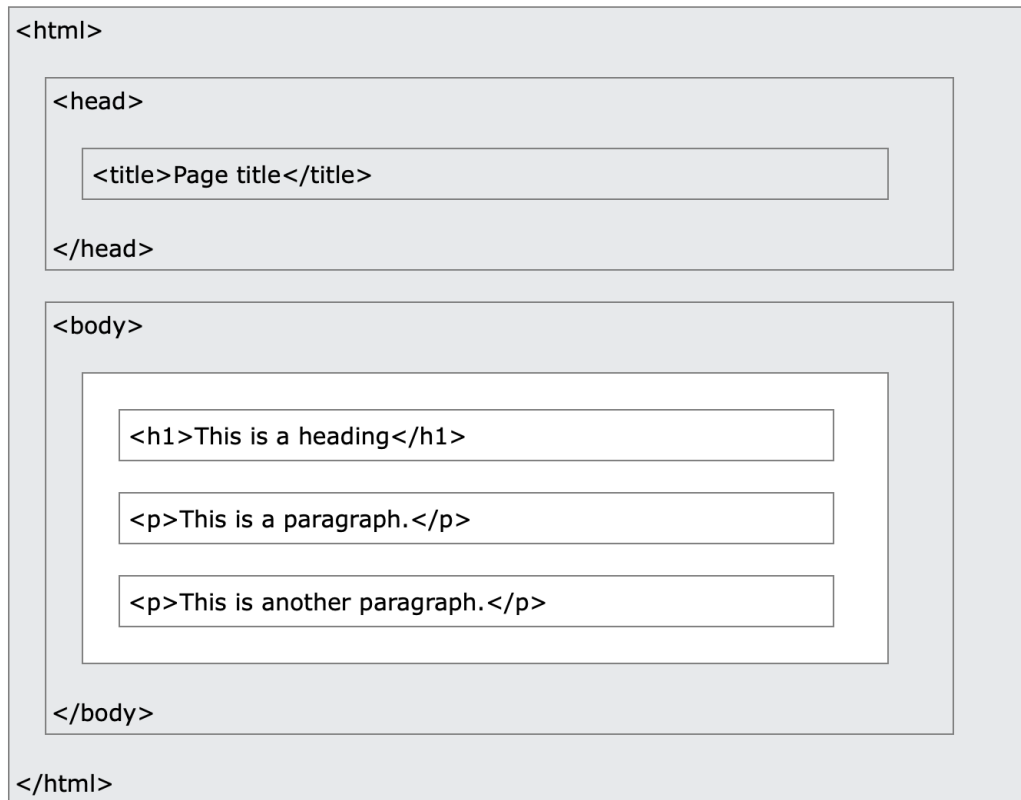
This is a paragraph.

HTML: Structure

An HTML element is a start tag, (optionally) content, and then (optionally) an end tag

Start tag	Element content	End tag
<h1>	My First Heading	</h1>
<p>	My first paragraph.	</p>
 	<i>none</i>	<i>none</i>

HTML: Structure



HTML: Headings

```
<h1>This is heading 1</h1>
<h2>This is heading 2</h2>
<h3>This is heading 3</h3>
```

This is heading 1

This is heading 2

This is heading 3

HTML: Learning

Ways to learn HTML:

- Modify an existing page
- Read tutorials on <https://www.w3schools.com>
- Use another markdown language (like qmd) that compiles to html, and study the result of the compilation
- Use dreamweaver or another professional piece of design software, inject your own html into aspects that aren't static designs

HTML: URLs

- Going back to the link, a URL is a “Uniform Resource Locators”
- Example: <http://example.com> Here the protocol is http (hyper text transport protocol) and the server is example.com
- Other examples of protocols: git, https, sftp

HTML: URLs

- A URL can also indicate a file on the server: <http://example.com/path/filename>
- Here “path/filename” can indicate a file on the server
- But, it doesn't have to:

<http://example.com/API?action=start-motor>

- Here there isn't (necessarily) a file on the server with the name “API?action=start-motor”. Instead, the question mark is indicating that parameters will be “passed” to a “program” called “API”

HTML: URLs

- Usernames and passwords can be embedded in URLs: <https://user:pa55w0rd@www.example.com/path/filename>
- The destination “port” can also be indicated: <https://user:pa55w0rd@www.example.com:8080/path/filename>
- The server listed in a URL can be specified in unicode through a special mapping from unicode to ASCII specified by the “Internationalized Resource Identifier” (IRI) standard.
- Example: [http:// .](http://xn-fsqu00a.xn-3lr804guic/) and <http://xn-fsqu00a.xn-3lr804guic/> are the same URL

The End

- Of new material
- Everything for the rest of today and next week will be review for the exams
- Please ask questions

Coding Exam Review

Key Ideas

- All material examinable, majority will be on the second half of the semester
- May be some concepts from before midterm
- Shorter than the practice exam (3 questions)

Preparing

- Do the practice exam, the format will be very similar
 - Download a dataset
 - Solve some short problems
 - Write functions, implement methods we have covered
 - Create and submit your pdf file

Submitting to Crowdmark

- Make sure you are comfortable with rendering your document!
- Do not display large data output, you will be penalised!
 - This is written in the instructions this time
- If you are asked to show a number, it must be in your pdf file for you to get points

Dealing with Errors

- You must submit a pdf, otherwise you will get 0
- We have covered how to deal with errors before
- Use `#| eval: false` in individual code chunks
 - Needs to be at the very start of chunk, spacing important
- Tip: Have a code chunk at the bottom of your template file with this, can copy and paste if needed

If all your code doesn't run

- Modify the YAML at the very top to not run any of your code

```
knitr:  
  opts_chunk:  
    eval: false
```

R Packages

- Will need to use packages, make sure to load them
- Any packages we saw in the class could be required
 - tidyverse
 - stringr
 - caret
 - class
 - RSql
 - DBI
 - etc.

Review of Key Topics

- Will spend the rest of the class reviewing each topic covered, types of questions you could get
- Make sure you can do all the labs, the midterm, you will be well prepared

Basic Programming

- Reading data into R
- Modifying data, accessing rows/columns, etc
- Basic plots of data
- Basic statistical functions (`mean`, `median`, `max`, etc)

Functions and Lists

- Be able to write simple functions in R
- Lists and how to access elements in lists

SQL and Databases

- Create and access databases with `RSQLite`
- Adding datasets, inserting rows
- Querying data from a database
 - `SELECT`, `FROM`, `WHERE`, `ORDER`, `AS`, `GROUP BY`, `LIKE`

Regression

- Fitting linear regression using `lm`
- Extract the $\hat{\alpha}, \hat{\beta}$ from the fitted model
- Basic use of `optim` for small problems (computing SSE)

APIs

- Understand how they work, be familiar with the ones we used in class

KNN and Prediction

- Split data into a training and test set
- Fit to training, predict on test
- Compute RMSPE
- Do this for linear regression and KNN

Classification and Cross Validation

- How to modify KNN to perform classification
 - Different function in different package
 - Compute accuracy instead of RMSPE
- Choose the value of K using cross-validation
 - How to perform this for new data

Clustering

- Implement k-means clustering and display output
- Know the key ideas in the algorithm
- How to modify the stopping criteria, change the number of random starts

Strings

- Will probably want to use `stringr`
- Will only use string functions, ideas covered in the slides

Likelihood and Mixture Models

- Be able to write an R function to compute the likelihood or log-likelihood
- Be able to plot the result for a range of values of the parameter
- Implement the EM algorithm (for very simple problem)